

EgoPed

Ego-Conditioned Pedestrian Body Motion Generation via Latent Diffusion

Generating realistic 3D full-body pedestrian motion conditioned on vehicle ego trajectory — for AV simulation.

Backbone: Motion Latent Diffusion (MLD), 263-D HumanML3D. Data: AVA + nuScenes + Waymo (11,954 samples).

Best system: EgoPed-IA — FID 2.880, a 40% improvement over pooled conditioning (unified evaluation).

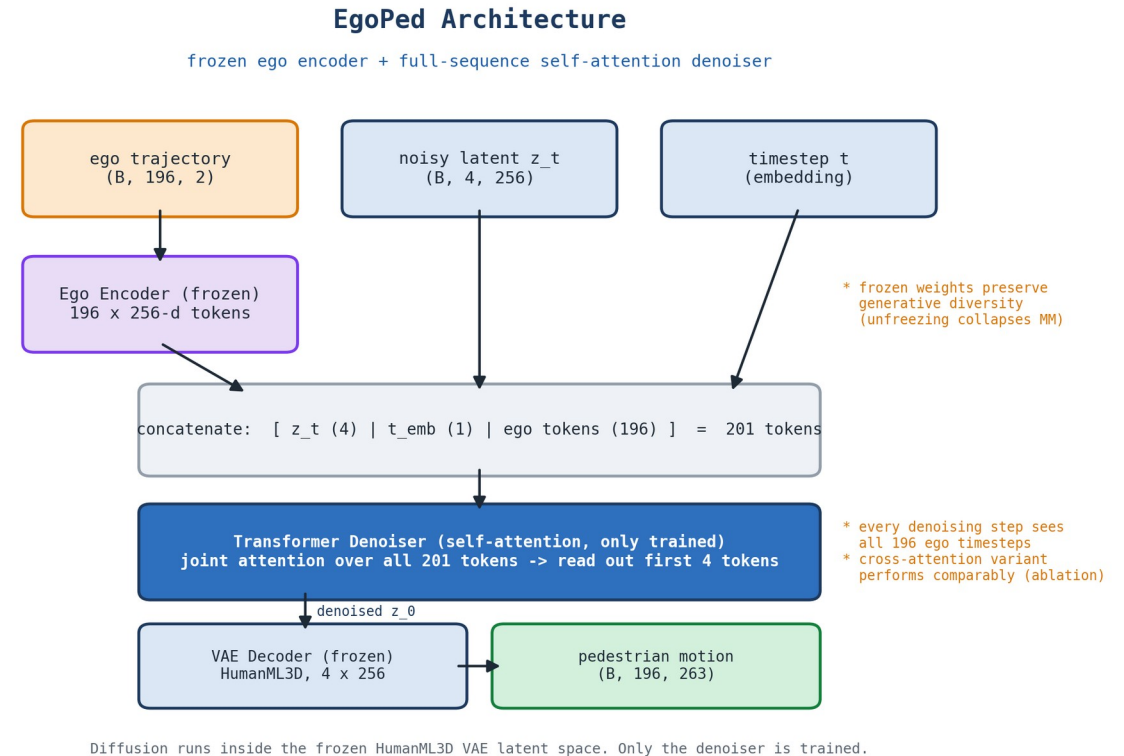
Validated on four independent axes: FID · held-out split · trajectory ADE/FDE · behavioral probe.

Problem & Motivation

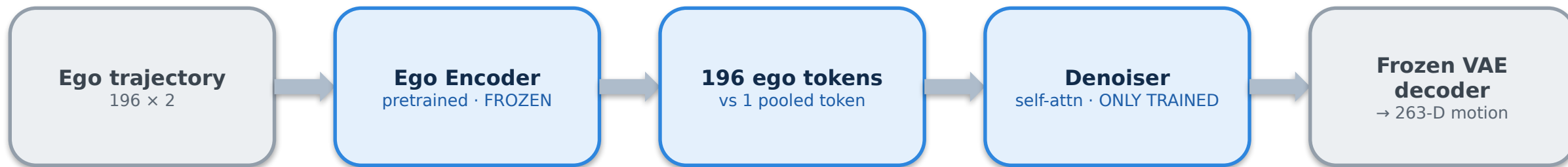
- AV simulation needs diverse, plausible pedestrian behavior to stress-test perception and planning.
- Existing 3D motion generators condition on text or action labels → context-agnostic: a crossing looks identical whether a car approaches at 50 km/h or is stopped.
- Real pedestrian behavior is coupled to the ego vehicle — speed, proximity and trajectory shape whether a person crosses, waits, accelerates or turns.
- Gap: ego-conditioned FULL-BODY synthesis is unexplored (prior work: 2D trajectories, or 3D bodies from text).
- Core question: how should the ego trajectory be injected into a latent-diffusion denoiser?

Approach

- MLD backbone: diffuse in a frozen HumanML3D VAE latent (263-D $\rightarrow$ 4 tokens $\times$ 256-D). Only the denoiser trains.
- Ego encoder: transformer maps the 196-step ego trajectory $\rightarrow$ conditioning tokens; contrastively pretrained, then FROZEN.
- Studied: pooled (1 mean-pooled token) vs full-sequence (all 196 per-timestep tokens).
- Full-sequence: 196 ego + timestep + 4 latent tokens jointly self-attended $\rightarrow$ per-timestep temporal resolution at every denoising step.
- CFG at inference; DDIM, 50 steps.



What We Changed vs. Base MLD



- Replaced text/CLIP conditioning with vehicle ego-trajectory conditioning (VAE, denoiser backbone and evaluators retained).
- New frozen ego encoder, contrastively pretrained against VAE motion latents.
- Changed conditioning granularity: all 196 per-timestep tokens vs one pooled token — the source of the gain.
- New interaction-aware data pipeline (scoring, weighted sampling, closest-approach cropping) — ablated, not assumed.

Dataset

- nuScenes + Waymo supply scale and diversity; AVA (our sensor vehicle) adds a small, deliberately STAGED high-interaction subset.
- Pipeline: MS3D annotation → per-camera 3D SMPL (adapted OmniRe) → stitch → 263-D HumanML3D → paired with ego odometry, pedestrian-centric frame.
- Interaction score = travel × (%frames within 5 m) × relative-bearing change → weighted sampling + closest-approach crop.

Source	Train	Val	Total
AVA (ours)	27	7	34
nuScenes	3,637	943	4,580
Waymo	5,876	1,464	7,340
Total	9,540	2,414	11,954

Splits are scene-disjoint. A scene-disjoint half of validation is further held out and never used for model or checkpoint selection.

Main Results

System	FID ↓	R@1	MM	Diversity
EgoPed-IA (ours, + interaction sampling)	2.880 ±0.084	0.323	3.37	5.67
EgoPed / H4 (ours, full-sequence)	3.392 ±0.179	0.548	3.96	5.79
H6 (unfrozen encoder)	5.079 ±0.034	0.352	2.62	6.06
H2 (pooled baseline)	5.620 ±0.117	0.665	3.71	5.82
H3 (latent-8 VAE)	6.677 ±0.135	0.663	3.37	5.89
Ground truth	—	—	—	5.50

- 40% FID improvement over pooled conditioning under a UNIFIED evaluation pipeline (identical eval-time processing for all systems).
- Verified conservative: the baseline additionally got interaction-aware training AND its best-of-3 seeds.
- Interaction-aware sampling improves the full-sequence model further, at 1/3 the training epochs.

Disentangling Architecture from Data

FID ↓	– interaction sampling	+ interaction sampling
H2 (pooled)	6.214 ±0.105	5.620 ±0.117
H4 (full-sequence)	3.392 ±0.179	2.880 ±0.084

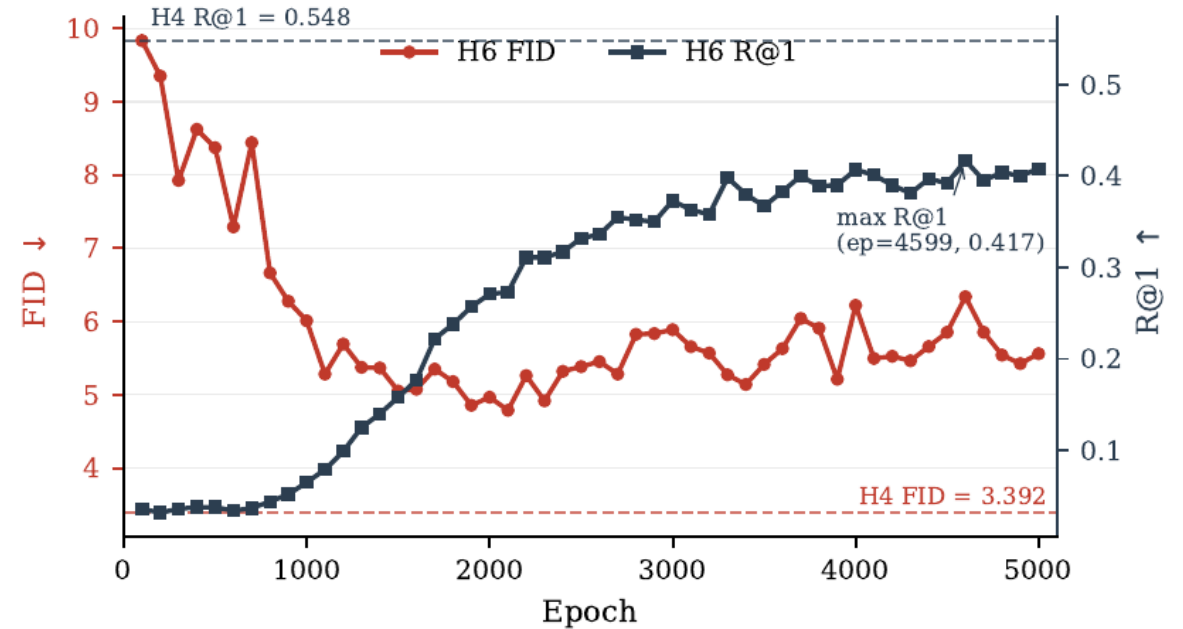
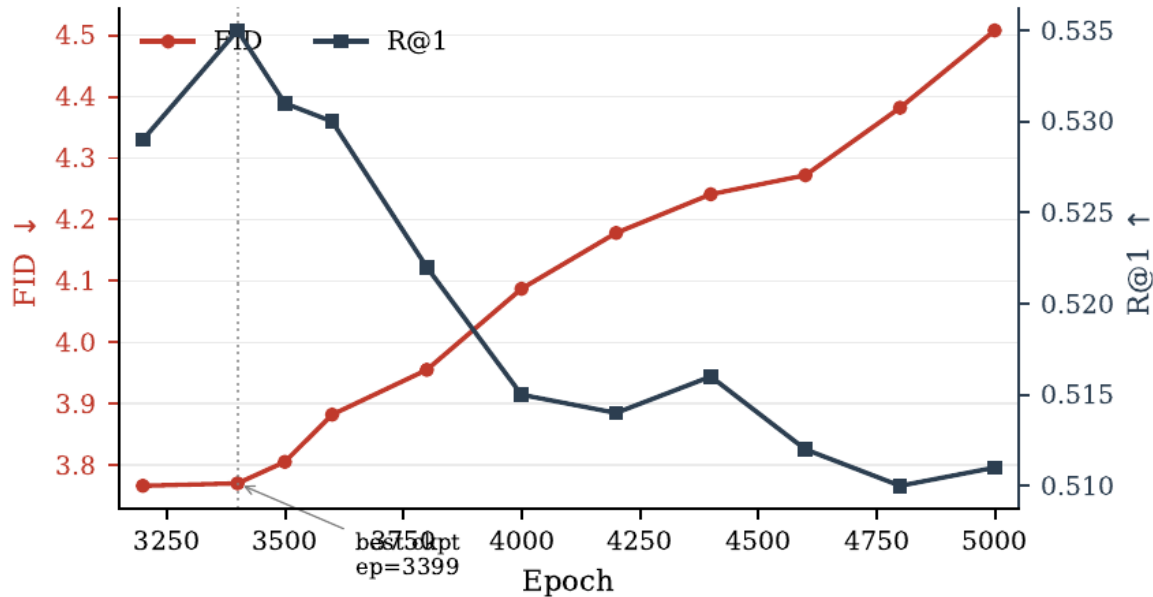
- Conditioning granularity dominates: ≈ 2.7 – 2.8 FID in BOTH columns — about $5\times$ the pipeline effect.
- The two interventions are approximately ADDITIVE (-0.59 pooled, -0.51 full-sequence): data curation helps, but it is not what drives the result.
- Why it matters: the original comparison mixed both factors. Separating them shows the headline was conservative, not inflated.

Ablations

Question	Finding	Verdict
Bigger VAE latent (8 vs 4)?	6.677 vs 5.620 FID	19% WORSE — reject
Unfreeze the ego encoder?	MM collapses 3.96 → 2.62	Freezing is critical
Skip encoder pretraining?	FID 4.968, R@1 = chance	≈ unconditional prior
Cross-attention instead of self-attn?	3.490 vs 3.338 (p = 0.33)	No difference
Different training seed?	3.39 / 4.72 / 4.80	Real variance — reported

- The encoder design is bracketed from BOTH sides: pretraining injects the ego information, freezing preserves generative diversity.
- The attention mechanism is NOT the source of the gain — full-sequence conditioning is (self-attention and cross-attention tie).
- Seed spread is reported honestly: single-seed point estimates overstate precision in this setting.

Checkpoint Selection & the Unfreeze Failure



- Left: FID does not improve monotonically — the final checkpoint is NOT the best checkpoint. Early stopping on validation FID is essential.
- Right: unfreezing the encoder never approaches the frozen model at any point in 5,000 epochs.

Evaluator-Independent Validation

Model	ADE ↓	FDE ↓	minADE ₅ ↓	minFDE ₅ ↓
EgoPed / H4 (full-sequence)	2.320	4.826	1.291	2.591
H2 (pooled)	2.434	4.990	1.482	2.979
Unconditional (ego zeroed)	2.880	5.876	1.228	2.454

- Root-trajectory displacement error uses NO learned evaluator and no model-specific embedding — directly comparable across architectures.
- H4 beats the pooled baseline on ALL FOUR metrics (paired over 1,190 held-out conditions; 3 of 4 significant) DESPITE its lower R-Precision.
- → the R@1 gap is an embedding-space artifact, not worse conditioning. Conditioning also demonstrably helps: 2.88 → 2.32 vs the unconditional prior.

Behavioral Validity

Model	Separation ↑	Entropy	Brier ↓
EgoPed / H4	0.333	0.238	0.149
H2 (pooled)	0.285	0.297	0.178
H6 (unfrozen)	0.364	0.178	0.178
Unconditional	0.021	0.533	0.199

- Probe: automatic stop/walk labels from root trajectories; compare each model's sampled $p(\text{stop} \mid \text{ego})$ against ground truth ($K = 5$ samples per condition).
- Sanity ✓ — the unconditional model is behaviorally ego-blind (separation ≈ 0), validating the probe.
- H4 couples behavioral decisions to the ego better than pooled, with the best calibration — a THIRD independent axis favouring full-sequence conditioning.
- H6 reveals its MultiModality collapse in behavior space: highest separation at the LOWEST entropy — it commits to one behavior per condition, overconfidently.

External Baselines: the Capability Ladder

Approach	FID ↓	Reading
EgoPed-IA (ours)	2.880	in-domain, ego-conditioned
Unconditional prior (ego zeroed)	5.18	conditioning is worth 34% FID
VAE latent interpolation	7.91	the learned prior matters
Pretrained text-to-motion MLD	14.07	off-the-shelf T2M does not transfer
Raw-space diffusion (MDM-style)	26.43	justifies the latent backbone
Ego→motion regressor (deterministic)	35.47	L2-optimal, yet unrealistic
Trajectory + static body	47.7–49.6	task needs articulated motion

- The regressor minimizes expected L2 by construction (ADE 1.80) but produces collapsed, unrealistic motion — while EgoPed's minADE₅ (1.29) beats it WITH realism.
- Pretrained MLD reported as the BEST of three prompts (14.07; the ladder spanned 14.07–24.84) — deliberately generous to the baseline.

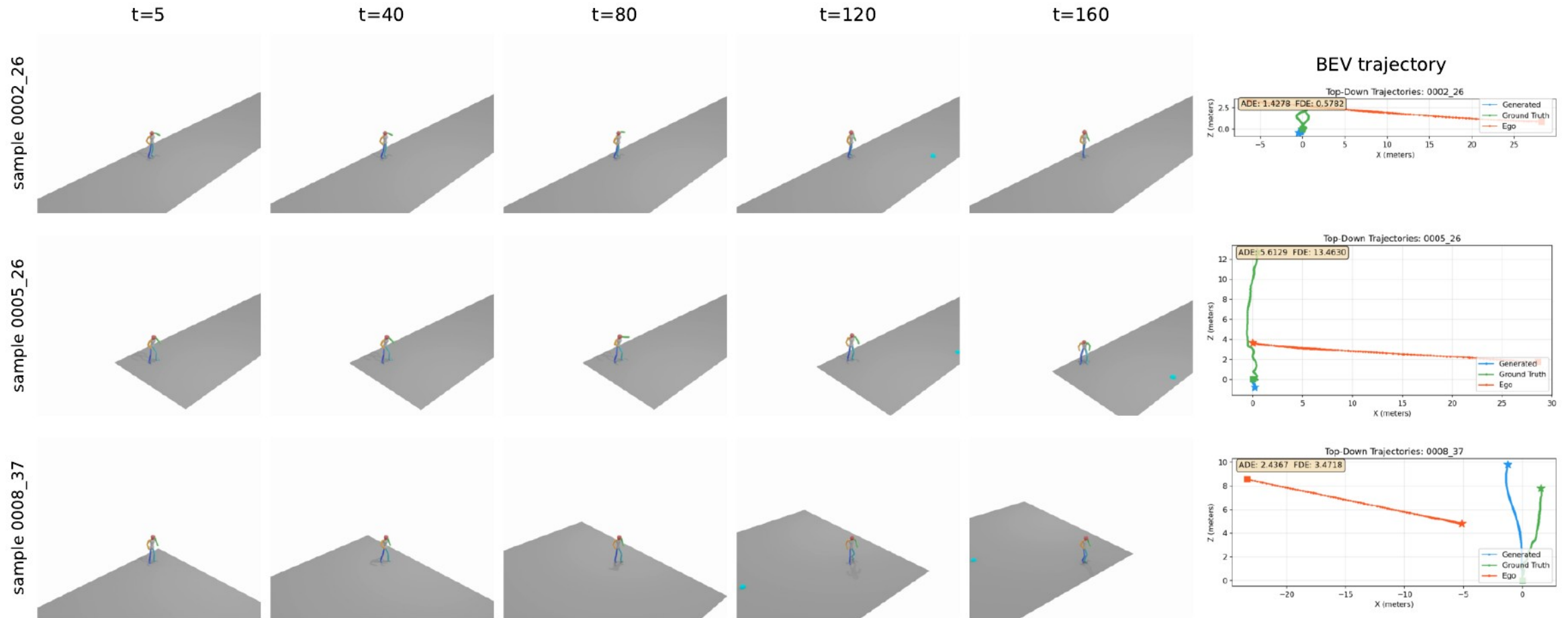
Retrieval (copying training motions) reaches FID 0.06 — a memorization sanity check, not a competitor.

How Much Conditioning Information Is Needed?

Conditioning signal	Information	FID ↓
Unconditional	0 bits	5.18 (prior) · training now
Body text (no vehicle)	2.99 bits	— (optional rung)
Ego text (vehicle verbalized)	6.09 bits	training now
Ego trajectory (ours)	continuous 196 × 2	2.880

- We synthesize captions deterministically from kinematics (no human annotation, no LLM): 14,172 captions, measured at 2.99 bits (body) and 6.09 bits (ego vocabulary).
- This turns the central claim into a QUANTITATIVE one: if the trajectory beats a ~6-bit linguistic description of the same interaction, fine-grained temporal conditioning carries information language cannot.
- Both runs are queued on the cluster; the comparison removes the domain gap that confounds the off-the-shelf text baseline.

Qualitative Results



Generated 3D body motion with bird's-eye trajectories: predicted (blue), ground truth (orange), ego vehicle (red).

Design Recipe & Contributions

- First system generating full 3D HumanML3D body motion conditioned on vehicle ego odometry (AVA + nuScenes + Waymo).
- Full-sequence temporal conditioning beats pooled conditioning by 40% FID — robust to CFG, to the attention mechanism, and amplified by interaction-aware sampling.
- The frozen, contrastively pretrained ego encoder is essential: pretraining injects the ego information, freezing preserves generative diversity.
- Recipe:
 - Pretrain the ego encoder against frozen VAE latents → freeze it → condition the denoiser on the FULL ego sequence.
 - Use CFG ≈ 10 ; select checkpoints on validation FID, never the final epoch.
 - Add interaction-aware sampling for a further gain at a third of the training cost.

Limitations & Next Steps

- Pose labels are estimated (OmniRe), not ground-truth 3D — label-quality audit outstanding.
- Conditioning uses 2D ego trajectory only; relative geometry, velocity profile and map context are unexplored.
- Seed sensitivity is real (FID 3.39 / 4.72 / 4.80); EgoPed-IA is reported from a single seed.
- No physics constraints (foot sliding), single benchmark, and no human perceptual study yet.
- In flight: unconditional and ego-text training runs to complete the information ladder.

EgoPed — full-sequence ego conditioning + frozen pretrained encoder: FID 2.880, validated on four independent axes.